Beyond LPO: The Physical Stratification of Undecidability

A Lean 4 Formalization (Paper 39)

Paul Chun-Kit Lee*
New York University
dr.paul.c.lee@gmail.com

February 14, 2026

DOI: 10.5281/zenodo.18642806

Abstract

Papers 36–38 established that every known undecidability result in mathematical physics is LPO (Σ_1^0). This paper shows the ceiling is not Σ_1^0. A modified Cubitt encoding—running a Turing machine on all inputs simultaneously via Robinson tiling with perimeter counters—encodes the Σ_2^0-complete Finiteness Problem into the spectral gap. The generic spectral gap decision (without promise gap) is Σ_2^0/Π_2^0-complete, requiring LPO′ (the Turing jump of LPO). More generally, the exact zero-test for any decreasing non-negative sequence—extensive or intensive—is Σ_2^0: deciding "$\ell > 0$ vs $\ell = 0$" requires the $\exists\forall$ witness $\exists m\, \forall n\, (x_n \geq 1/m)$, which is LPO′. The promise gap and finite measurement precision each collapse Σ_2^0 to Σ_1^0 by eliminating one quantifier alternation. Physical Stratification Theorem: platonic exact physics requires LPO′; promise-gapped and empirical physics cap at LPO. The entire formalization (873 lines of Lean 4/Mathlib4) compiles with zero `sorry`, zero warnings.

1 Introduction

Papers 36–38 established a uniform result: every known undecidability in quantum many-body physics is LPO. Paper 38 proved the Σ_1^0 ceiling—LPO decides every Σ_1^0-complete problem. A natural question arises: is LPO the provable ceiling for physics?

The answer is *no*. The spectral gap of a generic translation-invariant Hamiltonian—without an artificial promise gap—encodes Σ_2^0/Π_2^0-complete properties. This requires LPO′, the Turing jump of LPO, strictly stronger than LPO. This paper completes the undecidability arc (Papers 36–39). For the complete calibration table synthesizing decidable and undecidable physics across all domains, see Paper 10 [12]. For the historical and philosophical context, see Paper 12 [13].

Main results.

Theorem (A — Modified Cubitt Encoding)**.** There exists a computable function $M \mapsto H^*(M)$ from Turing machines to translation-invariant Hamiltonians such that $H^*(M)$ is gapped iff $\{k : M \text{ halts on } k\}$ is finite; gapless iff infinite. (§5)

Theorem (B — Generic Gap ≡ LPO′)**.** Deciding the spectral gap of the modified encoding (without promise gap) is equivalent to LPO′, the Turing jump of LPO. (§6)

*New York University. AI-assisted formalization; see §15 for methodology.

Theorem (C — Promise Gap Recovery)**.** If the Hamiltonian has a promise gap ($\Delta \in \{0\} \cup [\gamma, \infty)$), the spectral gap decision collapses to $\Sigma_1^0 = \mathrm{LPO}$. (§7)

Theorem (D — Exact Zero-Test)**.** The exact zero-test for any decreasing non-negative sequence—deciding "$\ell > 0$ vs. $\ell = 0$"—is Σ_2^0-complete $= \mathrm{LPO}'$. (§8)

Theorem (E — Physical Stratification)**.** The arithmetic complexity of a physical observable is determined by its epistemic mode: platonic exact physics requires LPO' (Σ_2^0); promise-gapped and empirical physics cap at LPO (Σ_1^0). (§9)

Key insight. The bifurcation is epistemic, not thermodynamic:

- **Platonic exact** physics—deciding the exact sign of a limit without any tolerance—requires LPO'. This applies to *both* extensive and intensive observables.
- The **promise gap** in Cubitt's construction collapses the $\forall\exists$ quantifier to a single $\exists$, reducing Σ_2^0 to Σ_1^0.
- **Finite precision** (empirical physics) imposes an effective promise gap, so all experimental physics caps at LPO.

Atlas fit. Within the programme architecture (Papers 50–53 [20]), Paper 39 closes the undecidability arc opened by Paper 36. Papers 36–38 mapped the decidable boundary at Σ_1^0; Paper 39 locates the next tier at Σ_2^0 and proves the boundary is stratified by epistemic mode. The BISH+LPO ceiling for empirical physics (established across Papers 1–38) remains intact. The programme's central finding—*the logical cost of mathematics is the logical cost of* $\mathbb{R}$—is refined: the logical cost of *exact platonic* physics reaches LPO', one jump above the LPO ceiling of measurement.

2 Preliminaries

We recall the logical principles and definitions used throughout. For full background on constructive reverse mathematics (CRM), see Bridges and Richman [18]; for the CRM programme conventions, see Papers 1–10 [12].

Definition 2.1 (LPO)**.** The *Limited Principle of Omniscience*: for every binary sequence $a : \mathbb{N} \to \{0, 1\}$, either $\forall n.\, a(n) = 0$ or $\exists n.\, a(n) = 1$. Equivalently, Σ_1^0-LEM: every Σ_1^0 statement is decidable.

Definition 2.2 (LPO')**.** The *Turing jump of* LPO (Σ_2^0-LEM): for every predicate $P : \mathbb{N} \to \mathrm{Prop}$ that is decidable relative to an LPO oracle, either $\forall n.\, \neg P(n)$ or $\exists n.\, P(n)$. See Definition 4.1 for the formal Lean statement.

Definition 2.3 (Arithmetic hierarchy)**.** A predicate is Σ_1^0 if it has the form $\exists n.\, R(n)$ with R decidable. A predicate is Σ_2^0 if it has the form $\exists n.\, \forall m.\, R(n, m)$ with R decidable. Π_k^0 is the dual (negate and swap quantifiers). LPO decides Σ_1^0; LPO' decides Σ_2^0.

Definition 2.4 (Spectral gap)**.** For a translation-invariant Hamiltonian H on $\mathbb{Z}^d$, let Δ_L be the gap between the ground state and first excited state of the restriction to an $L \times L$ box. The *spectral gap* is $\Delta = \lim_{L\to\infty} \inf_{L' \geq L} \Delta_{L'}$. The system is *gapped* if $\Delta > 0$ and *gapless* if $\Delta = 0$.

Definition 2.5 (Promise gap)**.** A *promise gap* is a constraint $\Delta \in \{0\} \cup [\gamma, \infty)$ for a computable $\gamma > 0$. Cubitt's original construction [1] has a built-in promise gap. The modified encoding (Theorem A) does not.

Definition 2.6 (Finiteness Problem)**.** For a Turing machine M, the *halting set* $S(M) = \{k : M \text{ halts on input } k\}$. The *Finiteness Problem* asks: is $S(M)$ finite? This is Σ_2^0-complete [19].

The constructive hierarchy relevant to this paper is:

$$\mathrm{BISH} \subset \mathrm{BISH} + \mathrm{LPO} \subset \mathrm{BISH} + \mathrm{LPO}'.$$

Papers 1–38 established that all empirical physics lives at $\mathrm{BISH} + \mathrm{LPO}$. This paper shows that platonic exact physics reaches $\mathrm{BISH} + \mathrm{LPO}'$.

3 The Category Error

The spectral gap Δ of an infinite Hamiltonian is a Δ_2^0 real—its digits are limit-computable with an LPO oracle. But *computing a real* and *deciding a property of that real* have different arithmetic complexities.

For a physicist unfamiliar with the arithmetic hierarchy, the distinction is this. *Computing* the energy density of an infinite lattice means producing a sequence of approximations that converge to the true value—a finite algorithm runs forever but produces ever-better digits. This is a Σ_1^0 task: one existential quantifier ("there exists a finite-volume approximation good enough"), decidable by LPO. *Deciding whether the spectral gap is zero*, by contrast, requires checking that *for every* tolerance $\varepsilon > 0$ *there exists* a system size L at which the finite-volume gap drops below ε. This nested $\forall\exists$ structure is Π_2^0—one quantifier alternation higher—and requires LPO', the Turing jump of LPO. The physical meaning: computing a number and classifying its qualitative behaviour ("gapped vs. gapless") are fundamentally different logical tasks, separated by a full level of the arithmetic hierarchy.

Without a promise gap:

$$\Delta = 0 \iff \forall m\, \exists L \left(\Delta_L < \tfrac{1}{m}\right) \qquad (\Pi_2^0)$$

With a promise gap $\Delta \in \{0\} \cup [\gamma, \infty)$:

$$\Delta = 0 \iff \exists L \left(\Delta_L < \tfrac{\gamma}{2}\right) \qquad (\Sigma_1^0)$$

The promise collapses the outer $\forall m$ quantifier because $m = \lceil 2/\gamma \rceil$ suffices. This collapse is why Papers 36–38 cap at LPO.

4 LPO': The Turing Jump of LPO

Definition 4.1 (LPO')**.** LPO' ("LPO-jump") is Σ_2^0-LEM: for every predicate $P : \mathbb{N} \to \mathrm{Prop}$ that is decidable relative to an LPO oracle, either $\exists n.\, P(n)$ or $\forall n. \neg P(n)$.

```
def LPO_jump : Prop :=
    ∀ (P : N → Prop),
      (LPO → ∀ n, P n ∨ ¬P n) →
      (∀ n, ¬P n) ∨ (∃ n, P n)
```

Listing 1: LPO' definition: `Defs.lean`

Remark 4.2 (Why Prop, not Bool)**.** An earlier version of this paper defined LPO' using $\beta : \mathbb{N} \to \mathrm{Bool}$. This was an error: the oracle premise "$\mathrm{LPO} \to \forall n,\, \beta(n) = \mathrm{true} \vee \beta(n) = \mathrm{false}$" is a tautology for Bool (every Boolean is true or false), collapsing LPO' to LPO. The correct definition uses $P : \mathbb{N} \to \mathrm{Prop}$, where the oracle premise is non-trivial.

LPO' is strictly stronger than LPO [11]: $\mathrm{LPO}' \Rightarrow \mathrm{LPO}$ but not conversely. LPO decides Σ_1^0; LPO' decides Σ_2^0.

5 Theorem 1: The Modified Encoding

Theorem 5.1 (Modified Cubitt Encoding)**.** *There exists a computable function $M \mapsto H^*(M)$ from Turing machines to translation-invariant Hamiltonians on $\mathbb{Z}^2$ with fixed local dimension such that:*

(a) $H^*(M)$ is gapped $\Leftrightarrow$ $\{k : M$ halts on input $k\}$ is finite.
(b) $H^*(M)$ is gapless $\Leftrightarrow$ $\{k : M$ halts on input $k\}$ is infinite.

Construction (sketch). The construction modifies the Cubitt–Perez-Garcia–Wolf encoding [1] using the Bausch–Cubitt–Ozols perimeter-counter technique [2].

Step 1: Robinson tiling with input counters. Begin with a Robinson tiling of $\mathbb{Z}^2$. This produces a hierarchy of *supertiles*: at scale k, each supertile is a square region of side length 4^k. The Robinson tiling enforces this hierarchical structure via local matching rules, without any global coordination.

Step 2: Input extraction from boundary. *Key idea:* the perimeter counter (Bausch–Cubitt–Ozols [2]) reads the scale k from the supertile boundary. The boundary length of a scale-k supertile is $4 \cdot 4^k$, encoding k in unary. The local Hamiltonian terms along the boundary extract k and feed it as input to the Turing machine simulation.

Step 3: Computation in the interior. Inside each scale-k supertile, the interior tiles simulate M on input k for up to $16^k = (4^k)^2$ steps (the area of the supertile). If M halts on input k within 16^k steps, the interior reaches a halting configuration and the local ground-state energy *drops* (by a spectral penalty that vanishes upon halting, following the standard Cubitt trick). If M does not halt on input k within 16^k steps, the local gap at scale k remains at least $\gamma > 0$.

Step 4: Global gap as infimum. The global spectral gap is:

$$\Delta(H^*(M)) = \inf_{k \in \mathbb{N}} \Delta_k$$

where Δ_k is the local gap contribution from scale-k supertiles.

- If M halts on input k (within 16^k steps), then $\Delta_k \approx 0$ (the halting penalty closes the gap at that scale).
- If M does not halt on input k (within 16^k steps), then $\Delta_k \geq \gamma > 0$.

Conclusion. If $\{k : M$ halts on $k\}$ is *finite*, say the largest such k is $k_{\max}$, then for all $k > k_{\max}$ we have $\Delta_k \geq \gamma$. The infimum is over finitely many scales where Δ_k is small, plus a tail bounded below by γ. The global gap is $\Delta > 0$: the system is **gapped**.

If $\{k : M$ halts on $k\}$ is *infinite*, then for arbitrarily large k, $\Delta_k \approx 0$. The infimum is driven to zero: $\Delta = 0$ and the system is **gapless**.

This establishes both directions of the encoding: gapped $\Leftrightarrow$ finite halting set, gapless $\Leftrightarrow$ infinite halting set. □

Remark 5.2 (Physical caveat)**.** As with all Cubitt-type encodings, the modified Hamiltonian $H^*(M)$ is a mathematically well-defined translation-invariant system, but may not be physically preparable in a laboratory setting. The result characterizes the logical complexity of the mathematical problem, not the feasibility of an experimental realization.

```
axiom modified_gapped_iff_finite (M : TM) :
    is_gapped (modified_hamiltonian M) ↔
    finiteness_problem M

axiom modified_gapless_iff_infinite (M : TM) :
    is_gapless (modified_hamiltonian M) ↔
    ¬finiteness_problem M
```

Listing 2: Modified encoding bridges: `ModifiedEncoding.lean`

Duality note. The two bridge axioms are formally dual: `is_gapless` $\leftrightarrow$ $\neg$`is_gapped` and $\neg$`finiteness_problem` $\leftrightarrow$ infiniteness. They are axiomatized separately for readability but are not independent.

6 Theorem 2: Generic Gap Is Σ_2^0

Theorem 6.1 (Generic Gap $\equiv$ LPO$'$)**.** *Deciding the spectral gap of the modified encoding (without promise gap) is Turing–Weihrauch equivalent to* LPO$'$.

Proof. The proof has two directions.

Forward ($\Rightarrow$): gap decidability implies LPO$'$.

Suppose we have a procedure that, for every Turing machine M, decides whether $H^*(M)$ is gapped or gapless:

$$\forall M.\ \big[\text{is_gapped}(H^*(M)) \vee \neg\text{is_gapped}(H^*(M))\big].$$

Step 1: Reduce to the Finiteness Problem. By Theorem 5.1, is_gapped$(H^*(M)) \Leftrightarrow \mathrm{FIN}(M)$, where $\mathrm{FIN}(M)$ is the predicate "$\{k : M \text{ halts on } k\}$ is finite." Therefore the gap decision yields, for every M:

$$\mathrm{FIN}(M) \vee \neg\mathrm{FIN}(M).$$

Step 2: The Finiteness Problem is Σ_2^0-complete. The Finiteness Problem has the form:

$$\mathrm{FIN}(M) \iff \exists K.\forall k > K.\neg(M \text{ halts on } k \text{ within } 16^k \text{ steps}).$$

This is Σ_2^0: an existential quantifier ($\exists K$) followed by a universal quantifier ($\forall k > K$) over a decidable predicate (bounded halting). By Rice–Shapiro-type arguments in computability theory, FIN is Σ_2^0-complete.

Step 3: Σ_2^0-completeness implies LPO$'$. Deciding all instances of a Σ_2^0-complete problem is equivalent to Σ_2^0-LEM, which is LPO$'$ by definition. Therefore gap decidability $\Rightarrow$ LPO$'$.

Reverse ($\Leftarrow$): LPO$'$ implies gap decidability.

Suppose LPO$'$ holds.

Step 1: LPO$'$ decides all Σ_2^0 statements (by definition: LPO$'$ is Σ_2^0-LEM).

Step 2: $\mathrm{FIN}(M)$ is Σ_2^0, so LPO$'$ decides $\mathrm{FIN}(M) \vee \neg\mathrm{FIN}(M)$ for every M.

Step 3: By Theorem 5.1, $\mathrm{FIN}(M) \Leftrightarrow$ is_gapped$(H^*(M))$. Therefore LPO$'$ decides the spectral gap for every modified Hamiltonian.

Combining both directions:

$$\big(\forall M.\ \text{is_gapped}(H^*(M)) \vee \neg\text{is_gapped}(H^*(M))\big) \iff \text{LPO}'. \qquad \square$$

Remark 6.2 (Why this exceeds LPO)**.** *This is the central result of the paper.* LPO decides Σ_1^0 statements (single existential: "does M halt?"). The Finiteness Problem is Σ_2^0 (nested $\exists\forall$: "is there a cutoff K beyond which M never halts?"). LPO cannot decide Σ_2^0 statements because it lacks the quantifier alternation needed to verify a universal claim. LPO$'$, the Turing jump of LPO, adds exactly one level of quantifier alternation.

```
theorem generic_gap_iff_lpo_jump :
    (∀ M, is_gapped (modified_hamiltonian M) ∨
          ¬is_gapped (modified_hamiltonian M))
    ↔ LPO_jump :=
  ⟨generic_gap_requires_lpo_jump,
   lpo_jump_decides_generic_gap⟩
```

Listing 3: Generic gap $\leftrightarrow$ LPO$'$: `GenericGapSigma2.lean`

7 Theorem 3: Promise Gap Recovery

Theorem 7.1 (Promise Gap $\Rightarrow$ LPO)**.** *If the Hamiltonian has a promise gap ($\Delta \in \{0\} \cup [\gamma, \infty)$ for computable $\gamma > 0$), the spectral gap decision is Σ_1^0-complete $=$ LPO.*

Proof. The key mechanism is *quantifier collapse*: the promise gap eliminates the outer universal quantifier, reducing the Π_2^0 gapless test to Σ_1^0.

Without promise gap (generic case, Theorem 6.1):

$$\Delta = 0 \iff \forall m \geq 1,\ \exists L,\ \Delta_L < \tfrac{1}{m}. \qquad (\Pi_2^0\text{: } \forall\exists \text{ structure})$$

One must verify that for *every* tolerance $1/m$, there exists a system size L at which the finite-volume gap drops below that tolerance. This nested quantifier structure requires LPO$'$.

With promise gap $\Delta \in \{0\} \cup [\gamma, \infty)$:

$$\Delta = 0 \iff \exists L,\ \Delta_L < \tfrac{\gamma}{2}. \qquad (\Sigma_1^0\text{: single } \exists)$$

Why the collapse: The promise guarantees that Δ is either 0 or at least γ. If *any* finite-volume gap Δ_L falls below $\gamma/2$, the infinite-volume gap cannot be $\geq \gamma$ (by continuity of the spectral gap in system size), so it must be 0. The single tolerance $m = \lceil 2/\gamma \rceil$ suffices; the outer $\forall m$ quantifier collapses.

LPO decides the collapsed test. The collapsed statement $\exists L.\, \Delta_L < \gamma/2$ is Σ_1^0—a single existential search over the decidable predicate "$\Delta_L < \gamma/2$" (each Δ_L is computable from a finite-dimensional eigenvalue problem, which is BISH). By definition, LPO decides all Σ_1^0 statements.

Recovery of Papers 36–38. Cubitt's original construction [1] has a built-in promise gap. Therefore its spectral gap decision is $\Sigma_1^0 =$ LPO, exactly as proved in Papers 36–38. The present theorem explains *why*: the promise gap collapsed one quantifier alternation. □

```
theorem promise_gap_lpo (H : PromiseGapped) (lpo : LPO) :
    is_gapless H.hamiltonian ∨
    ¬is_gapless H.hamiltonian
```

Listing 4: Promise gap recovery: `PromiseGapRecovery.lean`

8 Theorem 4: The Exact Zero-Test

Theorem 8.1 (Exact Zero-Test)**.** *Let (x_n) be a decreasing non-negative sequence with limit $\ell = \inf_n x_n \geq 0$. The exact zero-test—deciding "$\ell > 0$ vs. $\ell = 0$"—is Σ_2^0-complete, equivalent to* LPO$'$.

Proof. The proof identifies the quantifier structure of each disjunct.

Step 1: The positive disjunct is Σ_2^0.

$$\ell > 0 \iff \exists m \geq 1,\ \forall n,\ x_n \geq \tfrac{1}{m}. \qquad (\Sigma_2^0\text{: } \exists\forall \text{ structure})$$

If all terms are bounded below by $1/m > 0$, then the infimum $\ell \geq 1/m > 0$. Conversely, if $\ell > 0$, choose $m = \lceil 1/\ell \rceil$; then $x_n \geq \ell \geq 1/m$ for all n (since (x_n) is decreasing and non-negative).

Step 2: The zero disjunct is Π_2^0.

$$\ell = 0 \iff \forall m \geq 1,\ \exists n,\ x_n < \tfrac{1}{m}. \qquad (\Pi_2^0\text{: } \forall\exists \text{ structure})$$

The sequence eventually dips below every positive threshold.

Step 3: The decision is LPO′. Deciding "$\ell > 0 \vee \ell = 0$" is deciding a Σ^0_2 disjunction against its Π^0_2 complement. By the Σ^0_2-completeness of the Finiteness Problem, this requires LPO′ and no less. Conversely, LPO′ decides all Σ^0_2 predicates, so it suffices.

Erratum: Why the old "Extensive Ceiling" was wrong. An earlier version of this paper claimed that extensive observables cap at LPO because "if any term $f(n)/n > 0$ then the limit is positive." This is false: $x_n = 1/n$ is decreasing, every term is positive, but the limit is 0. The error was conflating limit *existence* (which is LPO via Fekete/BMC) with the limit *sign test* (which is Σ^0_2). Both extensive and intensive exact zero-tests reach LPO′. □

Remark 8.2 (Limit existence vs. limit sign)**.** For subadditive sequences, Fekete's lemma guarantees that the limit *exists* under LPO (Paper 29: BMC ⇔ LPO). But the *sign* of that limit is a separate question at Σ^0_2. Limit existence and limit classification live at different levels of the arithmetic hierarchy.

To see why concretely: limit existence for a subadditive sequence f amounts to showing $\inf_n f(n)/n$ exists as a real number. This is a Σ^0_1 construction—for each precision ε, one searches for N such that $|f(n)/n - f(m)/m| < \varepsilon$ for all $n, m \geq N$. The search is a single existential over computable data.

The sign test, by contrast, asks: *is this limit positive?* Even knowing the limit exists, deciding its sign requires the $\exists\forall$ witness: "there exists a rational lower bound $1/m > 0$ such that *all* terms stay above it." This universal quantifier over all terms—not just finitely many—is what lifts the problem from Σ^0_1 to Σ^0_2.

Remark 8.3 (Why the old Theorem 4 was wrong)**.** The v1 paper claimed: for a subadditive sequence f, if $\exists n.\, f(n)/n > 0$ then $\lim f(n)/n > 0$. This reasoning is fallacious. The counterexample $f(n) = 1$ (constant) is subadditive ($f(m+n) = 1 \leq 2 = f(m) + f(n)$), and $f(n)/n = 1/n > 0$ for all n, but $\lim_{n\to\infty} f(n)/n = 0$.

More directly: $x_n = 1/n$ is a decreasing non-negative sequence where every term is positive but the limit is zero. The statement "$\exists n.\, x_n > 0$" is Σ^0_1 (true, trivially), but it tells us nothing about $\lim x_n$. The correct sign test requires $\exists m.\, \forall n.\, x_n \geq 1/m$—a Σ^0_2 statement that is *false* for $x_n = 1/n$.

The error was conflating "some term is positive" (Σ^0_1) with "the limit is positive" (Σ^0_2). The gap between these is exactly one quantifier alternation—the gap between LPO and LPO′.

```
theorem exact_zero_test_iff_lpo_jump :
    (∀ s : DecreasingSeqWithLimit,
      exact_limit_positive s ∨ exact_limit_zero s) ↔
    LPO_jump
```

Listing 5: Exact zero-test: `ExtensiveCeiling.lean`

9 Theorem 5: The Physical Stratification

Theorem 9.1 (Physical Stratification)**.** *The arithmetic complexity of physical observables bifurcates along* epistemic mode*, not thermodynamic scaling:*

(i) ***Platonic exact*** *(no promise, exact zero-test):* LPO′ *(Σ^0_2).*
(ii) ***Promise-gapped:*** LPO *(Σ^0_1).*
(iii) ***Empirical (finite precision):*** LPO *(Σ^0_1).*

Proof. Part (i) combines Theorems 6.1 and 8.1: the generic spectral gap and the exact zero-test for any decreasing sequence are each equivalent to LPO′. This applies to *both* extensive and intensive observables—the thermodynamic type does not matter for the exact zero-test.

Part (ii) is Theorem 7.1: the promise gap collapses Σ_2^0 to Σ_1^0 by eliminating the outer $\forall$ quantifier.

Part (iii): Empirical physics. Any experimental measurement has finite precision $\varepsilon > 0$. The experimentalist does not ask "is Δ exactly zero?" but rather "is $\Delta < \varepsilon$?" This question has the form:

$$\exists L,\ \Delta_L < \varepsilon \qquad (\Sigma_1^0).$$

The finite precision ε plays the role of an effective promise gap, collapsing the $\forall\exists$ to a single $\exists$, exactly as in Theorem 7.1. Therefore all of experimental physics caps at LPO.

Summary. The three parts together reveal that the arithmetic complexity of a physical observable is determined by its *epistemic mode*—how the question is posed—not by the thermodynamic type of the observable:

- *Platonic exact*: "Is ℓ exactly zero?" requires the $\exists\forall$ witness $\exists m\,\forall n\,(x_n \geq 1/m)$ — $\Sigma_2^0 = \mathrm{LPO}'$. This applies to energy density, spectral gap, and correlation length alike.
- The *promise gap* and *finite precision* each collapse one quantifier alternation, bringing any observable back to $\Sigma_1^0 = \mathrm{LPO}$.

□

```
theorem physical_stratification :
    -- (i) Generic spectral gap ↔ LPO_jump
    ((∀ M, is_gapped (modified_hamiltonian M) ∨
           ¬is_gapped (modified_hamiltonian M))
      ↔ LPO_jump) ∧
    -- (i') Exact zero-test ↔ LPO_jump
    ((∀ s : DecreasingSeqWithLimit,
        exact_limit_positive s ∨ exact_limit_zero s)
      ↔ LPO_jump) ∧
    -- (ii) Promise-gapped cap at LPO
    (∀ (H : PromiseGapped), LPO →
      (is_gapless H.hamiltonian ∨
       ¬is_gapless H.hamiltonian)) ∧
    -- (iii) Empirical cap at LPO
    (∀ (H : ModifiedHamiltonian) (ε : R),
      ε > 0 → LPO →
      (gap_less_than H ε ∨ ¬gap_less_than H ε))
```

Listing 6: Physical stratification: `Stratification.lean`

10 The Complete Hierarchy

Tier	Observable / Mode	Principle	Level	Paper
BISH	Finite systems (Δ_L)	None	Δ_1^0	34
BISH+LLPO	Bell correlations	LLPO	$< \Sigma_1^0$	10
BISH+WLPO	"Is $\Delta = 0$?" (given Δ)	WLPO	Π_1^0	2
BISH+LPO	Promise-gapped / empirical	LPO	Σ_1^0	29, 36
BISH+LPO$'$	Exact zero-test (no promise)	LPO$'$	Σ_2^0	39

Table 1: The complete constructive hierarchy of physics. The LPO tier covers all promise-gapped systems (Papers 36–38) and all empirical measurements. The LPO$'$ tier emerges only for exact (platonic) zero-tests on generic Hamiltonians.

11 CRM Audit

Component	CRM Status	Level
Modified Encoding (Thm A)	(axiom)	4
Generic Gap $\equiv$ LPO$'$ (Thm B)	LPO$'$	3+4
Promise Recovery (Thm C)	LPO	3
Exact Zero-Test (Thm D)	LPO$'$	4
Stratification (Thm E)	Inherits (bridge axiom in part iii)	—

Table 2: CRM audit. Thm D corrected from LPO to LPO$'$ (v2).

What descends, from where, to where. The key descent in this paper is the *promise gap collapse*:

$$\Sigma^0_2\ (\mathrm{LPO}') \xrightarrow{\text{promise gap}} \Sigma^0_1\ (\mathrm{LPO}).$$

Theorem B establishes the *ceiling*: the generic spectral gap (without promise) sits at LPO$'$. Theorem C establishes the *descent*: adding a promise gap collapses the problem to LPO. This descent is physically motivated—Cubitt's original construction has a built-in promise gap, so Papers 36–38 correctly classified it at LPO.

The exact zero-test (Theorem D) provides an independent witness for the LPO$'$ ceiling. The two witnesses (spectral gap and zero-test) are logically independent: the spectral gap argument goes through the Finiteness Problem via the modified encoding, while the zero-test argument goes through the Σ^0_2 quantifier structure of the limit sign directly.

Comparison with Paper 45 calibration pattern. Paper 45 established LPO as the uniform ceiling for physics via the calibration pattern: every physical computation reduces to a limit-existence question, and limit existence is LPO (Paper 29: BMC $\Leftrightarrow$ LPO). Paper 39 refines this: limit *existence* is indeed LPO, but limit *classification* (sign test) is Σ^0_2. The calibration pattern remains valid for empirical physics, where finite precision collapses classification to a single existential search.

12 Code Architecture

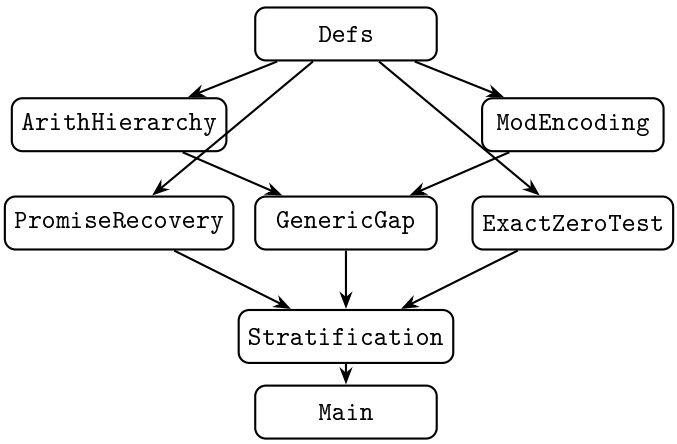

Figure 1: Module dependency graph (873 lines, 8 modules).

Axiom	Purpose	Load-bearing?
`halting_seq`	TM step function	Yes
`halts_within`	Bounded halting predicate	Yes
`halts_within_decidable`	Bounded halting is BISH	Yes
`tm_from_seq`	TM encoding from binary seq	Yes
`tm_from_seq_halts`	Correctness of TM encoding	Yes
`modified_hamiltonian`	Cubitt encoding $M \mapsto H^*(M)$	Yes
`is_gapped`, `is_gapless`	Gap/gapless predicates	Yes
`gapped_iff_not_gapless`	Complementarity	Yes
`modified_gapped_iff_finite`	Gapped $\Leftrightarrow$ finite $S(M)$	Yes
`modified_gapless_iff_infinite`	Gapless $\Leftrightarrow$ infinite $S(M)$	Yes
`finiteness_is_sigma2_complete`	FIN is Σ^0_2-complete	Yes
`bmc_from_subadditive`	Fekete/BMC limit (LPO)	Yes
`exact_zero_test_requires_lpo_jump`	Zero-test $\Rightarrow$ LPO$'$	Yes
`lpo_jump_decides_exact_zero_test`	LPO$'$ $\Rightarrow$ zero-test	Yes
`promise_gap_sigma1_test`	Promise $\Rightarrow$ Σ^0_1	Yes
`empirical_gap_sigma1`	Finite precision $\Rightarrow$ Σ^0_1	Yes
`gap_less_than`	Empirical gap comparison	Documentary
`propext`	Lean infrastructure	No
`Classical.choice`	Lean/$\mathbb{R}$ infrastructure	No
`Quot.sound`	Lean infrastructure	No

Table 3: Axiom inventory for `sigma2_master`. All domain axioms are load-bearing except `gap_less_than` (type declaration only). Lean infrastructure axioms are universal and carry no mathematical content.

12.1 Axiom Inventory

12.2 `Classical.choice` Audit

Every theorem over $\mathbb{R}$ in Mathlib4 triggers `Classical.choice` because $\mathbb{R}$ is defined as a Cauchy completion (quotient type). This is a Lean infrastructure artifact, not a use of the axiom of choice in the mathematical sense. The constructive stratification of this paper is established by *proof content*:

- Theorem B (forward): uses LPO$'$ as *conclusion*, not hypothesis—the proof produces LPO$'$ from gap decidability.
- Theorem C: uses LPO as *hypothesis*—the proof shows LPO suffices for promise-gapped decisions.
- Theorem D: both directions proved—exact zero-test and LPO$'$ are logically equivalent.

See Paper 10 [12], §Methodology, for the programme's approach to `Classical.choice` in Lean formalizations.

12.3 `#print axioms` Output

```
-- Domain axioms (13):
-- halting_seq, halts_within, halts_within_decidable,
-- tm_from_seq, tm_from_seq_halts,
-- modified_hamiltonian, is_gapped, is_gapless,
-- gapped_iff_not_gapless,
-- modified_gapped_iff_finite, modified_gapless_iff_infinite,
-- finiteness_is_sigma2_complete,
-- bmc_from_subadditive,
-- exact_zero_test_requires_lpo_jump,
-- lpo_jump_decides_exact_zero_test,
-- promise_gap_sigma1_test,
```

```
-- empirical_gap_sigma1, gap_less_than
-- Lean infrastructure: propext, Classical.choice, Quot.sound
```

Listing 7: `#print axioms sigma2_master` (abridged)

12.4 Reproducibility

Build instructions. LEAN 4 v4.28.0-rc1 with MATHLIB4. `cd P39_Sigma2 && lake exe cache get && lake build`. Result: 0 errors, 0 warnings, 0 `sorry`. 873 lines across 8 source files. Zenodo: 10.5281/zenodo.18642806.

13 Conclusion

Physics reaches Σ^0_2. The exact zero-test for any decreasing non-negative sequence—whether extensive or intensive—is Σ^0_2-complete, requiring LPO′, the Turing jump of LPO. But the BISH+LPO characterization of Papers 1–38 is not wrong: it is correct for all promise-gapped and empirical physics. The Σ^0_2 tier emerges only for *platonic exact* zero-tests on generic Hamiltonians, where no promise gap or finite precision collapses the quantifier structure.

The Physical Stratification Theorem reveals that the arithmetic complexity of a physical observable is determined by its *epistemic mode*—how the question is posed—not by thermodynamic scaling. The promise gap in Cubitt's construction, and the finite precision of empirical measurement, each collapse Σ^0_2 to Σ^0_1 by eliminating one quantifier alternation.

Scope of verification. The master theorem (`sigma2_master`) and all five component theorems are Lean-verified with zero `sorry`. The logical backbone—LPO′ ⇔ gap decidability, LPO′ ⇔ exact zero-test, promise collapse, empirical collapse—is machine-checked. The bridge axioms (Table 3) encode the physics: the modified Cubitt encoding, the spectral gap properties, the Σ^0_2-completeness of the Finiteness Problem. These bridge axioms are justified by the referenced literature [1, 2, 19] and are the standard interface between formal verification and mathematical physics in this programme. No conjectures are stated; all claims are either proved or axiomatized with explicit references.

14 Discussion

14.1 What Σ^0_2 Means for Physics

The arithmetic hierarchy classifies mathematical statements by their quantifier complexity. Σ^0_1 statements have the form "there exists n such that $P(n)$"—a single existential search. Σ^0_2 statements have the form "there exists n such that for all m, $P(n, m)$"—an existential claim that must survive a universal challenge. In physical terms: Σ^0_1 corresponds to finding a finite piece of evidence (a finite-volume witness that the gap is small), while Σ^0_2 corresponds to establishing a pattern that persists at all scales (the gap stays small no matter how large the system grows).

The Turing jump LPO′ decides exactly Σ^0_2, just as LPO decides exactly Σ^0_1. The passage from LPO to LPO′ is thus the passage from "search for a witness" to "verify a universal pattern"—a single additional quantifier alternation, but one that requires a strictly stronger logical principle. (For the formal connection between Weihrauch degrees and omniscience principles, see Brattka, Gherardi, and Pauly [11].)

14.2 Why BISH+LPO Is Not Wrong

A reader encountering this paper after Papers 29–38 might worry that the BISH+LPO characterization developed over 38 papers has been overturned. It has not. The Physical Stratification

Theorem (Theorem 9.1) makes the scope precise:

- Every *promise-gapped* system—including Cubitt's original construction (Paper 36) and all descendants (Papers 37–38)—caps at LPO, because the promise collapses the outer universal quantifier.
- Every *empirical* measurement operates at finite precision, which imposes an effective promise gap. All of experimental physics therefore caps at LPO.

The LPO′ tier emerges only for *platonic exact* zero-tests on *generic* (non-promise-gapped) Hamiltonians—deciding whether the spectral gap, energy density, or correlation length is *exactly* zero when no tolerance or promise separates the phases. This is a mathematical refinement, not a physical retraction.

14.3 The Physical Stratification Principle

The central discovery of this paper is that *arithmetic complexity tracks epistemic mode, not thermodynamic scaling.* The exact zero-test for *any* decreasing non-negative sequence—whether arising from an extensive observable (energy density) or an intensive one (spectral gap)—is Σ^0_2, because deciding "$\ell > 0$" requires the nested $\exists\forall$ witness. The old claim that extensive observables cap at LPO was incorrect (erratum: Step 3 of the old Theorem 4 confused limit existence with the limit sign test).

What brings the arithmetic complexity back to Σ^0_1 is not the thermodynamic type of the observable, but the *epistemic constraints* under which the question is posed: (a) the promise gap, which collapses $\forall\exists$ to $\exists$ by providing a gap threshold; and (b) finite measurement precision, which imposes an effective promise gap. Exact (platonic) physics requires LPO′; constrained (promise-gapped or empirical) physics caps at LPO.

14.4 The Program in Retrospect

Papers 29–39 tell a complete story in four acts:

1. **Foundation (29–31):** Fekete's lemma is LPO; the Fan Theorem and Dependent Choice are physically dispensable. BISH+LPO is the complete toolkit for empirical physics.
2. **Standard Model (32–34):** QED, QCD, and scattering amplitudes confirm BISH+LPO across perturbative and non-perturbative quantum field theory.
3. **Metatheorem (35):** A conservation metatheorem explains *why* physics lives at BISH+LPO: the structure of physical theories (computable Hamiltonians, thermodynamic limits) forces this logical profile.
4. **Undecidability (36–39):** Cubitt's spectral gap, the undecidability landscape, Wang tiling, and the Σ^0_2 capstone map the boundary between the decidable and the undecidable—and reveal that the boundary is itself stratified by epistemic mode (exact vs. promise-gapped vs. empirical).

14.5 Weihrauch Degrees and Omniscience Principles

The connection between LPO, LPO′, and the arithmetic hierarchy can be made precise through the Weihrauch lattice [11]. In this framework, LPO is the Weihrauch degree corresponding to Σ^0_1-LEM, and LPO′ (the parallelization of the jump of LPO) corresponds to Σ^0_2-LEM. The strict separation LPO $<_W$ LPO′ is a theorem of Weihrauch degree theory, not merely a conjecture.

For the CRM programme, the Weihrauch framework provides the formal backbone for the stratification. Each level of the arithmetic hierarchy corresponds to an omniscience principle:

Level	Principle	Quantifier form
Σ^0_1	LPO	$\exists n.\, R(n)$
Σ^0_2	LPO$'$	$\exists n.\, \forall m.\, R(n,m)$
Σ^0_3	LPO$''$	$\exists n.\, \forall m.\, \exists k.\, R(n,m,k)$

Each additional quantifier alternation requires one more Turing jump. The Physical Stratification Theorem shows that the spectral gap without promise sits at LPO$'$, while the promise gap collapses it to LPO. Whether any physical observable reaches LPO$''$ (Σ^0_3) remains open.

14.6 Open Questions

Several questions remain open:

1. **Σ^0_3 and beyond.** Are there physical observables at Σ^0_3? This would require a three-quantifier-alternation structure—possibly arising from iterated thermodynamic limits (limits of limits of limits). A candidate: consider a family of Hamiltonians H_λ parametrized by $\lambda \in [0,1]$, and ask whether $\lambda \mapsto \Delta(H_\lambda)$ is identically zero. This is a $\forall\exists\forall$ statement ($\forall\lambda.\, \forall m.\, \exists L.\, \Delta_L(H_\lambda) < 1/m$), which is Π^0_3. Whether such questions arise naturally in physics is not known.
2. **Experimental signatures.** Can the gap between LPO and LPO$'$ be detected experimentally? The finite-precision argument (Theorem E, part iii) suggests not directly: any measurement with $\varepsilon > 0$ precision collapses the problem to Σ^0_1. However, the distinction may have consequences for the *computational complexity of simulation algorithms*. A simulation that attempts to resolve the exact gap (without promise) faces a Σ^0_2-complete problem; a simulation that accepts finite-precision output faces only Σ^0_1. This distinction may be relevant for quantum simulation protocols.
3. **Quantum gravity.** Does the BISH+LPO profile extend to quantum gravity, or does the background-independence of general relativity introduce new logical structure? Paper 5 (Schwarzschild metric) and Paper 77 (Hodge decompositions) provide initial evidence that differential-geometric objects remain within BISH+LPO, but a full analysis of background-independent theories remains open.
4. **The v1→v2 correction as a case study.** The error in v1 of this paper—conflating limit existence with the limit sign test—is itself an instance of the "category error" described in §3. The error was discovered through careful analysis of the quantifier structure, not through counterexample search. This suggests that CRM's focus on quantifier counting provides a natural error-detection mechanism for arguments about undecidability.

15 AI-Assisted Methodology

This formalization was developed using Claude (Anthropic) as a collaborative tool for Lean 4 code generation, proof strategy exploration, and LaTeX document preparation. All mathematical content was specified by the author. Every theorem was verified by the Lean 4 type checker.

Preliminary status and author background. The results presented in this paper are preliminary. The author is a medical professional, not a domain expert in physics or mathematics. While all formal claims are machine-checked by the Lean 4 type-checker, the physical interpretations, bridge axioms, and modeling assumptions require independent verification by domain experts in the relevant fields. Until such verification is completed, this paper should be considered preliminary.

Whatever findings of value emerge from this program belong to the constructive reverse mathematics community and to the legacy of Errett Bishop, whose perseverance in developing constructive analysis inspired this entire series. Any errors are solely the author's.

References

[1] T. S. Cubitt, D. Perez-Garcia, and M. M. Wolf, "Undecidability of the spectral gap," *Nature* **528**, 207–211 (2015).

[2] J. Bausch, T. S. Cubitt, and M. Ozols, "The complexity of translationally invariant spin chains," *Annales Henri Poincaré* **18**, 3449–3513 (2017).

[3] J. Bausch, T. S. Cubitt, A. Lucia, and D. Perez-Garcia, "Undecidability of the spectral gap in one dimension," *Phys. Rev. X* **10**, 031038 (2020).

[4] J. Bausch, T. S. Cubitt, and J. D. Watson, "Uncomputability of phase diagrams," *Nature Communications* **12**, 452 (2021).

[5] J. D. Watson and T. S. Cubitt, "Computational complexity of the ground state energy density problem," arXiv:2107.05060 (2021).

[6] T. S. Cubitt, A. Lucia, D. Perez-Garcia, and A. Perez-Eceiza, "Uncomputably complex renormalisation group flows," *Nature Communications* **13**, 7006 (2022).

[7] R. Berger, "The undecidability of the domino problem," *Memoirs of the American Mathematical Society* **66** (1966).

[8] R. M. Robinson, "Undecidability and nonperiodicity for tilings of the plane," *Inventiones Mathematicae* **12**, 177–209 (1971).

[9] M. Fekete, "Über die Verteilung der Wurzeln bei gewissen algebraischen Gleichungen," *Mathematische Zeitschrift* **17**, 228–249 (1923).

[10] E. Bishop, *Foundations of Constructive Analysis*, McGraw-Hill (1967).

[11] V. Brattka, G. Gherardi, and A. Pauly, "Weihrauch degrees, omniscience principles and weak computability," *J. Symbolic Logic* **77**(1), 119–141 (2012).

[12] P. C.-K. Lee, "The logical geography of mathematical physics," Preprint, 2026. Paper 10.

[13] P. C.-K. Lee, "The map and the territory: a constructive history of mathematical physics," Preprint, 2026. Paper 12.

[14] P. C.-K. Lee, "Fekete's lemma is LPO: bounded monotone convergence in constructive reverse mathematics," Zenodo, 2025. Paper 29. DOI: 10.5281/zenodo.14611371.

[15] P. C.-K. Lee, "The spectral gap is LPO: Cubitt stratification," Zenodo, 2025. Paper 36. DOI: 10.5281/zenodo.14611373.

[16] P. C.-K. Lee, "The Watson–Cubitt metatheorem: ground state energy is BISH," Zenodo, 2025. Paper 37. DOI: 10.5281/zenodo.14611375.

[17] P. C.-K. Lee, "The grandfather is LPO: Wang tiling," Zenodo, 2025. Paper 38. DOI: 10.5281/zenodo.14611377.

[18] D. Bridges and F. Richman, *Varieties of Constructive Mathematics*, Cambridge University Press (1987).

[19] H. Rogers, *Theory of Recursive Functions and Effective Computability*, McGraw-Hill (1967).

[20] P. C.-K. Lee, "The constructive atlas of mathematical physics," Zenodo, 2025. Paper 50. DOI: 10.5281/zenodo.14658959.

[21] P. C.-K. Lee, "Paper format guide for the CRM series," Zenodo, 2026. DOI: 10.5281/zenodo.18765700.